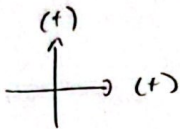


THINKING/ INQUIRY Clearly show ALL of your work for full credit. All questions on this test must be answered / solved using energy and momentum concepts where possible. Answers that do not use energy and momentum concepts will not receive full marks.

Round your final answer to 3 sd.

1. During a football game, a fullback of mass 108 kg, running at a velocity of 9.1 m/s [E], is tackled head-on by a defensive back of mass 91 kg, running at a velocity of 6.3 m/s [W]. What is the final velocity of the pair that are joined together just after the collision? (3 marks)

$$m_1 = 108 \text{ kg}$$
$$v_{1i} = 9.1 \text{ m/s [E]}$$



$$m_2 = 91 \text{ kg}$$

$$v_{2i} = -6.3 \text{ m/s [E]}$$

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

$$v_f = \frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2}$$

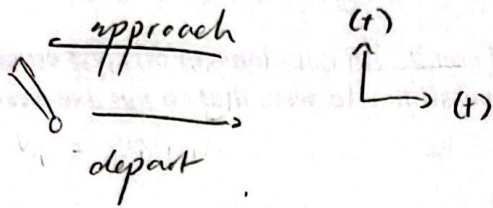
$$= \frac{(108 \text{ kg})(9.1 \text{ m/s [E]}) + (91 \text{ kg})(-6.3 \text{ m/s [E]})}{108 \text{ kg} + 91 \text{ kg}}$$

$$= 2.0577 \text{ m/s [E]}$$

the final velocity of the pair is 2.06 m/s [E]

3

2. A baseball with a mass of 0.14 kg has an initial velocity of 38 m/s as it approaches a bat. The bat applies an average force that is much larger than the weight of the ball, and the ball departs from the bat with a final velocity of 58 m/s. If the time of contact was 1.6 ms, find the average force exerted on the ball by the bat. (4 marks)



$$m = 0.14 \text{ kg}$$

$$v_i = -38 \text{ m/s [E]}$$

$$v_f = 58 \text{ m/s [E]}$$

$$\Delta t = 1.6 \text{ ms} \rightarrow 1.6 \times 10^{-3} \text{ s}$$

$$F \Delta t = m \Delta v$$

$$F = \frac{m \Delta v}{\Delta t}$$

$$= \frac{0.14 \text{ kg} (58 \text{ m/s [E]} - (-38 \text{ m/s [E]}))}{1.6 \times 10^{-3} \text{ s}}$$

$$= 8400 \text{ N}$$

the force exerted by the ball is 8400 N F

4

$$\frac{-12.867 \text{ kg m/s} + 10.392 \text{ kg m/s})^2}{52.8 \text{ N/m}})^{\frac{1}{2}} \quad \text{Crystal be}$$

$$\frac{5 \text{ kg} (6.1241 \text{ kg m/s})}{52.8 \text{ N/m}}$$

$$= 0.7615 \text{ m}$$

the compression of the spring was $|0.762 \text{ m}|$

1.432m

m_1 is held in front of a compressed spring. When m_1 is released it travels forward until it hits a stationary mass, the collision is a glancing collision as shown in the diagram. Using the following values, calculate the compression of the spring. There is no friction. (8 marks)

$$m_1 = 5.00 \text{ kg}$$

$$m_2 = 2.00 \text{ kg}$$

$$v_{2i}/v_{2f} = 6.00 \text{ m/s}$$

$$k = 52.8 \text{ N/Kg}$$

$$\theta = 25.0^\circ$$

$$\phi = 30.0^\circ$$

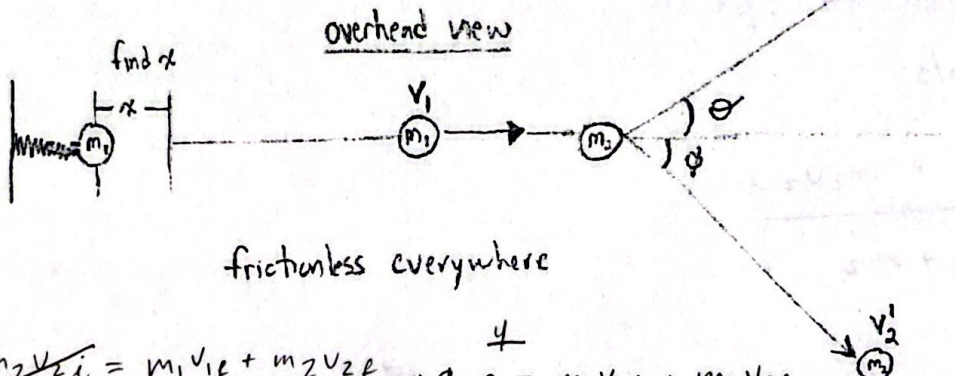
No numbers until the very end!

$$v_{2i} = 0 \text{ m/s}$$

$$v_{1i} = ?$$

$$v_{1f} = ?$$

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$$



frictionless everywhere

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f} \cos \phi$$

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} \cos \theta + m_2 v_{2f} \sin \phi$$

$$v_{1f} = -\frac{m_2 v_{2f} \sin \phi}{m_1 \sin \theta}$$

$$m_1 v_{1i} + m_2 v_{2i} = m_1 \left(\frac{-m_2 v_{2f} \sin \phi}{m_1 \sin \theta} \right) \cos \theta + m_2 v_{2f} \cos \phi$$

$$v_{1i} = \frac{-m_2 v_{2f} \sin \phi \cos \theta}{\sin \theta} + m_2 v_{2f} \cos \phi$$

* FINISHED ON SCRAP, FINAL

ANSWER = 0.762 m

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$$

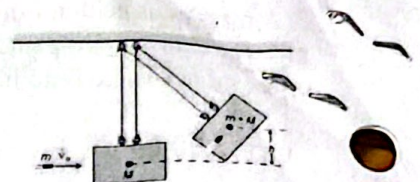
$$v_i = 4.652 \text{ m/s}$$

$$x = \sqrt{\frac{mv^2}{k}}$$

$$x = \sqrt{\frac{m_1 \left[\left(\frac{-m_2 v_{2f} \sin \phi \cos \theta}{\sin \theta} \right) + m_2 v_{2f} \cos \phi \right]^2}{k}}$$

$$x = \sqrt{\frac{(5 \text{ kg}) \left(\frac{(-2 \text{ kg})(6 \text{ m/s})(\sin 30^\circ)(\cos 25^\circ)}{\sin 25^\circ} + (2 \text{ kg})(6 \text{ m/s})(\cos 30^\circ) \right)^2}{52.8 \text{ N/m}}}$$

4. A ballistic pendulum can be used to measure the speed of a bullet. The one in the diagram consists of a stationary 3.20 kg block of wood suspended from a wire. A bullet with a mass of 2.0×10^{-2} kg is fired with a speed of 3000 km/h into the block. The block and bullet combination swing as illustrated. At what height does the block-bullet combination swing up to? Include conservation statements where necessary. (5 marks)



bullet

wood

$$\frac{1}{2}mv^2 = mgh$$

$$m = 2.0 \times 10^{-2} \text{ kg} \quad m = 3.20 \text{ kg}$$

$$v_{1i} = 3000 \text{ km/h} \quad v_{1i} = 0 \text{ m/s}$$

$$= 833.3 \text{ m/s}$$

$$v_f = \frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2}$$

$$\frac{1}{2}mv^2 = mgh$$

$$\frac{1}{2} \left(\frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2} \right)^2 = gh$$

$$h = \frac{0.5 \left(\frac{(2.0 \times 10^{-2} \text{ kg})(833.3 \text{ m/s}) + (3.2 \text{ kg})(0 \text{ m/s})}{2.0 \times 10^{-2} \text{ kg} + 3.2 \text{ kg}} \right)^2}{9.81 \text{ m/s}^2}$$

$$h = \frac{0.5 (5.1759 \text{ m/s})^2}{9.81 \text{ m/s}^2}$$

$$= 1.3659 \text{ m}$$

the height the bullet-block combo swings till is 1.37 m up

5